Deliverable 1
Structural Aliasing Analysis in Time Series Forecasting using Chronos

Matteo Boniotti, Gianluca Brignoli and Federico Sabbadini *

May 27, 2026

This document must address the following key questions:

- How will the project achieve its communication objectives?
- Descrizione dettagliata dei diversi punti da affrontare
- Quali sono i risultati attesi?
- Quali metodologie utilizzare per costruire dati sintetici?
- ?Analisi rappresentatività? (perdita di informazione e problematiche embedding (Probing MDL????))
- Analisi sui dati reali e risutlati attesi (Bayesian statistics)
- Implicazioni in termini di sicurezza sul dominio

This investigation formalizes the phenomenon of *structural aliasing* within patch-based time-series foundation models, focusing specifically on Amazon's Chronos-Bolt framework.

Purpose and Scope

This project deploys a highly localized, layer-resolved evaluation blueprint across five distinct stages of the transformer architecture, probing immediately after individual encoder and decoder blocks, and directly following the final direct quantile output projection layer.

By mapping representations across these specific layers, the project isolates where task-relevant signal attributes are preserved or compromised under architectural synchronization constraints. The investigation evaluates the model using a custom hyperparameter configuration, where patch size is held constant at $P = 16$, and the stride is systematically modified within the range of $S \in [8, 16]$ to explore the explicit impact of patch geometry on aliasing effects.

Ultimately, this study aims to provide a comprehensive understanding of the relationships between raw signal characteristics and the latent representations learned by the model, systematically identifying the critical links between sampling frequency (f_s), patch size (P), and stride (S) in order to propose actionable, principled architectural solutions to mitigate the phenomenon.

To ground this abstract architectural evaluation, the study utilizes a specific physical domain as empirical reference layer. This reference setting models the solar energy produced by a photovoltaic (PV) array attached to an internal load, a local storage battery, and the national electricity grid. The explicit operational objective within this system is to evaluate production variations over short-term future horizons (the next minutes or hours) so that the system can dynamically modify the internal load, attach and detach the battery unit, with the aim of maintaining internal grid stability and enabling precise anomaly detection.

During the preparation of this work, the authors used AI-based tools, including GPT-5.5, GPT-5.4, Gemini 3.1 Pro, and Sonnet 4.6, to support language refinement and improve clarity. All generated or suggested content was carefully reviewed, revised where necessary, and validated by the authors, who assume full responsibility for the final content of this work.

Within this infrastructure, a control AI agent is responsible for executing real-time tasks: first, **dynamic load-balancing**, which requires orchestrating the instantaneous routing of generated photovoltaic energy to meet shifting internal consumption demands; second, **battery dispatch**, which manages the continuous connection, isolation, or disconnection of the physical storage system to act as a buffer for intermittent generation; third, **state-monitoring and active anomaly detection**, parsing system-wide telemetry to differentiate between standard ambient fluctuations and hard sensor corruptions or hardware faults; and fourth, **internal grid stability maintenance**, acting as a localized governor to minimize power distribution imbalances before they escape to the national grid interface.

Intended Audience and Impact

This analysis is directly intended for researchers and practitioners utilizing large language models within the Chronos family for time-series analysis, representation learning, and forecasting, as well as individuals interested in characterizing the fundamental limitations, structural boundaries, and latent capabilities of these architectures. The core objective is to challenge the baseline assumption that pre-trained temporal foundation models inherently preserve frequency signal attributes when processing continuous data tracks, provided they satisfy classical sampling theorems.

This study is also explicitly directed toward researchers, engineers, and practitioners working in the field of energy management and domains directly related to this described infrastructure. For professionals operating within this space, the analysis exposes how algorithmic data-deletion anomalies can propagate through automated distribution pipelines. Evaluating how hidden token-space representation collapses can deceive a load-balancing AI agent into executing misaligned control choices, this work provides critical insights required to build high-reliability, fault-tolerant deployment models for smart-grid management systems.

Significance and Impact

This subject is of fundamental importance because it explore an unexamined systemic vulnerability native to patch-based temporal foundation architectures. By systematically characterizing the conditions under which structural aliasing emerges, this work provides critical insights into the fundamental limitations of these models, directly informing the design of more robust architectures and training paradigms.

The impact of this project extends beyond theoretical understanding, as it has direct implications for the reliability and safety of real-world applications that depend on accurate time-series forecasting and representation learning, particularly in high-stakes domains such as energy management.